

25. Sharpening of Image Using Gradient Masking

Aim

To perform **image sharpening using Gradient Masking** by detecting and enhancing the edges of an image using gradient masks.

Masks Used

Horizontal Gradient Mask (G_x):

Vertical Gradient Mask (G_y):

Procedure

1. Read the input image.
2. Convert the image into grayscale.
3. Apply the **G_x** mask to detect horizontal edges.
4. Apply the **G_y** mask to detect vertical edges.
5. Calculate the gradient magnitude from both edge images.
6. Add the gradient information to the original image.
7. Normalize the pixel values to the range 0–255.
8. Display the original and sharpened images.
9. Save the sharpened image.

Program

```
import cv2

import numpy as np

# Read input image

img = cv2.imread("input.jpg")
```

if img is None:

```
    print("Error: Image not found")
```

```
    exit()
```

Convert to grayscale

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

Gradient masks

```
Gx = np.array([
```

```
    [-1, -2, -1],
```

```
    [ 0,  0,  0],
```

```
    [ 1,  2,  1]
```

```
], dtype=np.float32)
```

```
Gy = np.array([
```

```
    [-1,  0,  1],
```

```
    [-2,  0,  2],
```

```
    [-1,  0,  1]
```

```
], dtype=np.float32)
```

Apply gradient masks

```
gx = cv2.filter2D(gray, cv2.CV_32F, Gx)
```

```
gy = cv2.filter2D(gray, cv2.CV_32F, Gy)
```

```
# Calculate gradient magnitude

gradient = cv2.magnitude(gx, gy)


# Normalize gradient

gradient = cv2.normalize(gradient, None, 0, 255, cv2.NORM_MINMAX)


# Convert to uint8

gradient = gradient.astype(np.uint8)


# Sharpen image

sharpened = cv2.add(gray, gradient)


# Display images

cv2.imshow("Original Image", gray)

cv2.imshow("Gradient Mask", gradient)

cv2.imshow("Sharpened Image", sharpened)


# Save output

cv2.imwrite("gradient_sharpened.jpg", sharpened)

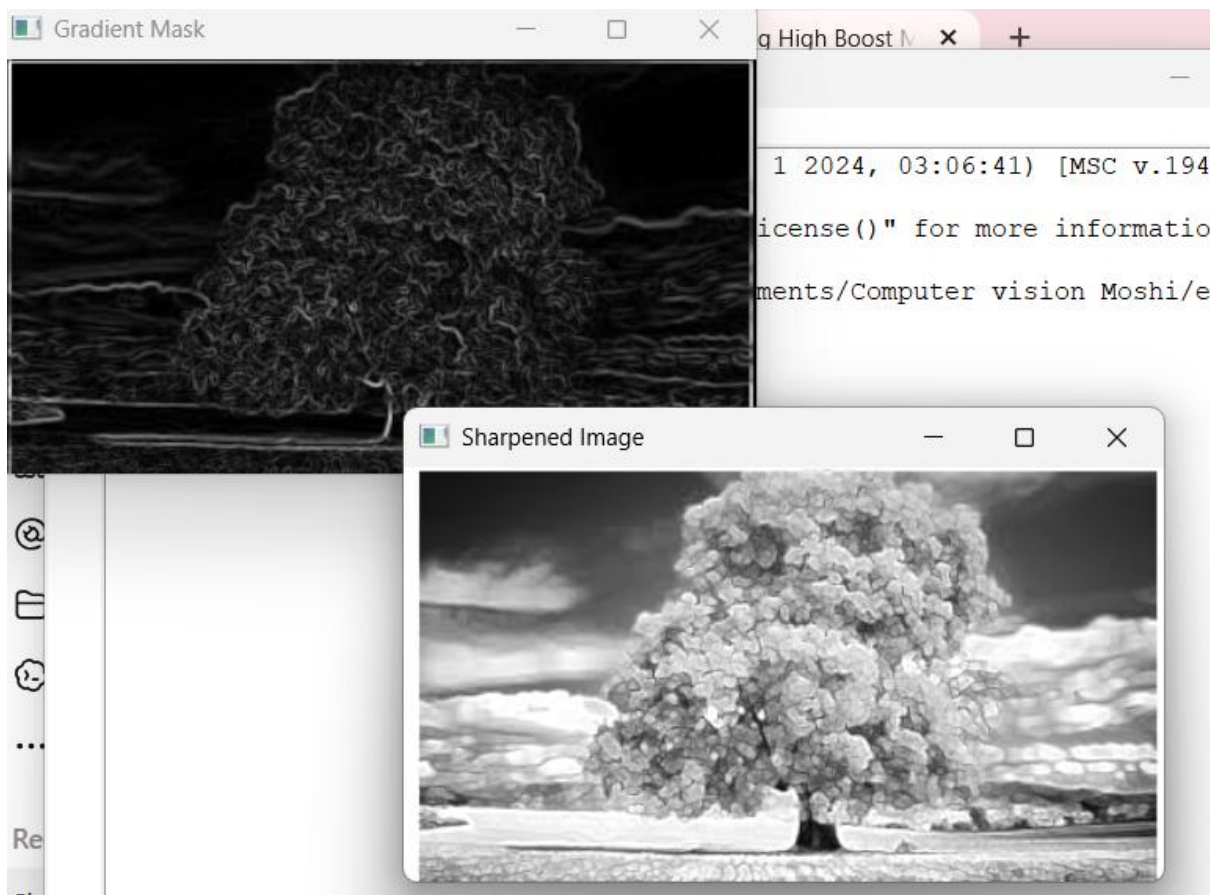

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Input



Output



Result

The image was successfully sharpened using **Gradient Masking**. The horizontal and vertical gradients were calculated and added to the original image, resulting in enhanced edges and improved image sharpness.